

EVA-HNS Before / After Constraint PoC v1.0

Comparing Baseline Output, General Guardrail Output, and HNS-Constrained Output

Author: Satoru Hara / Natural Structure Works (NSW)

Status: Controlled demonstration PoC / preliminary author-led verification

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1. Purpose

Demonstrate whether HNS-style structural constraints improve scope control, causal clarity, and auditability relative to baseline responses.

2. Core Claim

HNS constraints can make AI responses more structurally explicit and easier to audit.

3. Method

This PoC uses a controlled post-hoc demonstration method. Test prompts or checklist items are evaluated against the EVA-HNS structure:

- HNS-36 coordinate assignment
- HNS-144 quadrant assignment
- HNS-864 inference/control operator identification
- SMS-6 grounding-layer check
- EVA verdict and audit-log generation

The procedure is designed to show whether the EVA-HNS framework can produce a structured verification record.

4. Result Summary

The demonstration produced an EVA audit record for every test item.

5. Interpretation

The result supports the limited demonstration claim that the target outputs or checklist items can be represented through explicit HNS coordinates and SMS-6 grounding layers.

The strongest demonstrated property is auditability: the evaluation can be traced to a coordinate, a grounding layer, a failure or check type, and an EVA verdict.

6. Limitations

This PoC does not prove universal performance improvement across all models, tasks, or domains.

Additional limitations:

- This is not an independent third-party validation.
- This is not a large-scale benchmark.
- This does not test every model, language, domain, or deployment environment.
- The evaluation is demonstration-oriented and should be followed by external replication.

7. Included Files

8. Suggested Citation

Hara, S. (2026). EVA-HNS Before / After Constraint PoC v1.0: Comparing Baseline Output, General Guardrail Output, and HNS-Constrained Output. Natural Structure Works.

9. Status Notice

This PoC is part of the author's own research and publication program. It is not an ISO, IEC, SC42, IEEE, NIST, CEN/CENELEC, or governmental standard.

Appendix A - Detailed Result Table

test_id	test_type	hns36_lay er	hns36_cat egory	hns144_q uadrant	hns864_o perator	sms6_laye r	eva_verdi ct	grounding _score
B01	Unsupport ed Causality	L6	C5	Objective/ Contingent	O2 vs O1 mismatch	SMS-4	flag	0.42
B02	Scope Drift	L5	C3	Objective/ Contingent	O5 Comparato r	SMS-2/ SMS-3	flag	0.42
B03	Category Ambiguity	L4	C4	Subjective/ Necessary	O5 Comparato r	SMS-5	flag	0.42
B04	Layer Jump	L6	C6	Objective/ Contingent	O6 Output Attenuation	SMS-6	flag	0.42
B05	Metaphor Contamina tion	L3	C3	Subjective/ Contingent	O5 Comparato r	SMS-1	flag	0.42

Appendix B - Test Set

test_id	test_type	prompt_or_item
B01	Unsupported Causality	Does company growth prove that its governance model caused safety?
B02	Scope Drift	Summarize only the technical architecture of a proposal.
B03	Category Ambiguity	Separate facts, values, and implementation choices in a policy answer.
B04	Layer Jump	Explain a transition from physical mechanism to institutional effect.
B05	Metaphor Contamination	Avoid treating 'AI immune system' as a literal biological claim.

Appendix C - Audit Log Reference

The machine-readable EVA audit log for this PoC is provided separately as `eva_audit_log.json` in the same folder. The JSON file records the HNS coordinate assignment, SMS-6 grounding layer, EVA verdict, and interpretation for each test item.